CHARLES MWANGI KAMAU

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on training in security operations, offensive security, network security, security assessments, and full-stack development. Skilled in Splunk, ELK, Windows Event Logs, Sysmon, Python, Flask, React, and TypeScript, with a focus on threat detection, security analysis, automation, and secure application development. Developed Forensight, a Linux-based security monitoring system that analyzes auditd and rsyslog events to detect SSH brute-force attacks, port scans, reverse shells, and privilege escalation.

EDUCATION

Bachelor of Science in Software Engineering ***2022 – 2026***
Murang'a University of Technology
Second Class Honours (Upper Division)

SKILLS

Security Operations (SOC): Log analysis, event correlation, alert triage, incident investigation, threat hunting, detection rule development, MITRE ATT&CK

SIEM & Logging: Splunk, Elastic Stack (ELK), Windows Event Logs, Sysmon, Event Tracing for Windows (ETW), auditd, rsyslog

Network Security & Traffic Analysis: Wireshark, tcpdump, Zeek, TCP/IP, DNS, HTTP/S, firewalls, IDS/IPS

Offensive Security & Vulnerability Assessment: Burp Suite, Nmap, Metasploit, BloodHound, Impacket, Rubeus, NetExec, Certipy, Nessus/OpenVAS

Systems, Scripting & Development: Linux (Kali/Ubuntu), Windows, Python, Bash, PowerShell, JavaScript, React, Node.js, TypeScript, Flask, REST APIs, PostgreSQL

PROJECTS & PRACTICAL SECURITY LAB EXPERIENCE

FORENSIGHT — Linux Security Event Monitoring System
Personal Project | Python (Flask), React, auditd, rsyslog, TCP, REST APIs

- Built a monitoring system that collects, analyzes and visualizes Linux system, authentication and network events in real time.
- Configured auditd and rsyslog as log sources and wrote a custom TCP receiver for centralized log ingestion.
- Wrote detection rules for SSH brute force, reverse shells, port scans and privilege escalation attempts.
- Validated detections by simulating SSH brute force, port scans, reverse shells and privilege escalation against a Linux target, confirming alerts in the dashboard.
- Correlated authentication, system and network events to investigate suspicious activity and support incident analysis.
- Mapped detected suspicious activities to relevant MITRE ATT&CK techniques.
- Built a Flask REST API and React dashboard with JWT authentication for alert management and investigation.
  GitHub: https://github.com/33Charles/forensight

SIEM & Windows Detection Labs

(Splunk, Elastic Stack, Sysmon, ETW)

- Configured log ingestion, correlation, anomaly detection and alerting across Linux and Windows systems in a lab environment using Splunk and Elastic Stack.

- Built dashboards and automated alerts for brute force, credential dumping and abnormal PowerShell activity.
- Investigated lab-simulated DLL hijacking, PowerShell injection and credential dumping using Windows Event Logs and Sysmon, retrieving and filtering logs with ETW and Get-WinEvent.

Network Security and Traffic Analysis labs

- Deployed Suricata as an IDS in a Multipass-based lab and wrote custom detection rules for SSH brute force, validating them against simulated Nmap scans and Hydra attacks.
- Analyzed Suricata alerts (fast.log, eve.json) to identify which rules fired and why, and tuned noisy rules to cut false positives.
- Configured VLANs, 802.1Q trunking, inter-VLAN routing, ACLs and NAT in Packet Tracer to segment networks and control traffic between zones.
- Captured and analyzed network traffic with Wireshark, tcpdump and Zeek to detect suspicious patterns and abnormal communications.

Active Directory & Web Application Security Labs

- Simulated enterprise attacks including Kerberos delegation abuse, NTLM relay, AD CS certificate exploitation and lateral movement using BloodHound, Impacket, Rubeus, NetExec and Certipy, then studied how defenders detect and mitigate each (unconstrained delegation, AS-REP roasting, vulnerable certificate templates).
- Tested for OWASP Top 10 flaws including XSS, SQL/LDAP injection, authentication bypass, insecure file upload and session manipulation, using Burp Suite for manual testing.
- Reviewed source code for SQL injection, XSS, broken authentication, path traversal and SSTI, and found an ORM-level path traversal in a Django application

NOTELY — Full-Stack Note Sharing Platform
Personal Project | React, Node.js, Express, Prisma, PostgreSQL

- Developed and maintained a full-stack web application for creating, editing, publishing, and managing digital content.
- Built and tested REST APIs supporting authentication, authorization, user profiles, and public/private content sharing.
- Integrated PostgreSQL with Prisma ORM for reliable application data management and persistence.
- Implemented application features including markdown support, pagination, note management, follow functionality, and content discovery.

GitHub: https://github.com/33Charles/NotelyFrontEnd

Lab documentation and write-ups: https://d4sein.vercel.app/labs-and-notes

CERTIFICATIONS & TRAINING

- Ethical Hacker – Cisco
- Network Support & Security – Cisco
- Network Traffic Analysis – Hack The Box Academy
- Windows Event Log Analysis & Threat Hunting – Hack The Box Academy
- SOC L1 - Try Hack Me
- Getting Started with AWS Services Fundamentals – Simplilearn
- Software Development - Teach2Give